Example 5.4 Find the missing angle.

(i) (ii)

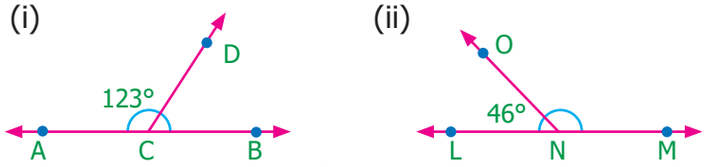


Fig.5.7

Solution

(i) Since the angles are linear pair, $\angle ACD + \angle BCD = 180°$

$123° + \angle BCD = 180°$

Subtracting 123° on both sides

$123° + \angle BCD - 123° = 180° - 123°$

$\angle BCD = 57°$

(ii) Since the angles are linear pair, $\angle LNO + \angle MNO = 180°$

$46° + \angle MNO = 180°$

Subtracting 46° on both sides

$46° + \angle MNO - 46° = 180° - 46°$

$\angle MNO = 134°$

Example 5.5 Two angles are in the ratio 3:2. If they are linear pair, find them.

Solution

Let the angles be $3x$ and $2x$

Since they are linear pair of angles, their sum is 180°.

Therefore, $3x+2x = 180°$

$5x = 180°$

$x = \frac{180°}{5}$

$x = 36°$

The angles are $3x = 3\times36 = 108°$

$2x = 2\times36 = 72°$

More on linear pairs

Amudhan asked his teacher what would happen if he drew a ray in between a linear pair of angles? The teacher told him to draw it. Amudhan drew the ray as shown in Fig.5.8.

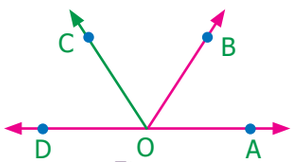


Fig.5.8

Teacher asked Amudhan, "what can you say about the angles $\angle AOB$ and $\angle BOC$?". He said that they are adjacent angles. Also it is true that $\angle AOB + \angle BOC = \angle AOC$.

The teacher also asked about the pair of angles $\angle AOC$ and $\angle COD$. He replied that they are linear pair. Therefore, their sum is 180° i.e. $\angle AOC + \angle COD = 180°$.

Combining these two results we get $\angle AOB + \angle BOC + \angle COD = 180°$.

Thus, the sum of all the angles formed at a point on a straight line is 180°.

7th_Maths_T1_EM_Chp5.indd 89 20-12-2021 21:29:29